Choline: Fact Sheet for Consumers (original version)

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What is choline and what does it do?

Choline is a nutrient that is found in many foods. Your brain and nervous system need it to regulate memory, mood, muscle control, and other functions. You also need choline to form the membranes that surround your body's cells. You can make a small amount of choline in your liver, but most of the choline in your body comes from the food you eat.

How much choline do I need?

The amount of choline you need each day depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	125 mg
Infants 7–12 months	150 mg
Children 1–3 years	200 mg
Children 4–8 years	250 mg
Children 9–13 years	375 mg
Teen boys 14–18 years	550 mg
Teen girls 14–18 years	400 mg
Men 19+ years	550 mg
Women 19+ years	425 mg
Pregnant teens and women	450 mg
Breastfeeding teens and women	550 mg

What foods provide choline?

Many foods contain choline. You can get recommended amounts of choline by eating a variety of foods, including the following:

- Meat, eggs, poultry, fish, and dairy products
- Potatoes and cruciferous vegetables such as brussels sprouts, broccoli, and cauliflower
- Some types of beans, nuts, seeds, and whole grains

What kinds of choline dietary supplements are available?

Some multivitamin/mineral supplements contain choline, often in the form of choline bitartrate, phosphatidylcholine, or lecithin. Dietary supplements that contain only choline are also available.

Am I getting enough choline?

The diets of most people in the United States provide less than the recommended amounts of choline. Even when choline intakes from both food and dietary supplements are combined, total choline intakes for most people are below recommended amounts.

Certain groups of people are more likely than others to have trouble getting enough choline:

- Pregnant women
- People with certain genetic conditions
- People who are being fed intravenously

What happens if I don't get enough choline?

Although most people in the United States don't get recommended amounts of choline, few people have symptoms of choline deficiency. One reason might be that our bodies can make some choline. However, if a person's choline levels drop too low, he or she can experience muscle and liver damage as well as deposits of fat in the liver (a condition called nonalcoholic fatty liver disease [NAFLD] that can damage the liver).

What are some effects of choline on health?

Scientists are studying choline to better understand how it affects health. Here are some examples of what this research has shown.

Cardiovascular disease

Some research shows that getting enough choline might help keep the heart and blood vessels healthy, partly by reducing blood pressure. Other research suggests that higher amounts of choline might increase cardiovascular disease risk. More research is needed to understand whether getting more choline from the diet and supplements might raise or lower the risk of cardiovascular disease.

Neurological disorders

Some studies have found a link between higher intakes of choline (and higher blood levels of choline) and better cognitive function (such as verbal and visual memory). However, other studies have shown that choline supplements do not improve cognition in healthy adults or in patients with Alzheimer's disease, Parkinson's disease dementia, or other memory problems. More research is needed to understand the relationship between choline intakes and cognitive function as well as to find out whether choline supplements offer any benefit to patients with dementia.

Nonalcoholic fatty liver disease

There may be a link between low intakes of choline and the risk of developing NAFLD. NAFLD is a condition in which fat builds up in the liver of people who do not drink excessive amounts of alcohol. It is a common liver disorder, especially in people who are overweight or have obesity. Getting enough choline is necessary for proper liver function and to prevent NAFLD. However, more research is needed to better understand how choline might help prevent or treat NAFLD.

Can choline be harmful?

Getting too much choline can cause a fishy body odor, vomiting, heavy sweating and salivation, low blood pressure, and liver damage. Some research also suggests that high amounts of choline may increase the risk of heart disease.

The daily upper limits for choline include intakes from all sources—food, beverages, and supplements—and are listed below.

Life Stage	Upper Limit
Birth to 12 months	Not established
Children 1–3 years	1,000 mg
Children 4–8 years	1,000 mg
Children 9–13 years	2,000 mg
Teens 14–18 years	3,000 mg
Adults	3,500 mg

Does choline interact with medications or other dietary supplements?

Choline is not known to interact with any medications.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as choline.

Choline and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the *Dietary Guidelines for Americans*.

Where can I find out more about choline?

- For general information about choline
  - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on Choline
- For more information on food sources of choline
  - USDA's FoodData Central
  - Nutrient List for choline (listed by choline content), USDA
- For more advice on choosing dietary supplements
  - ODS Frequently Asked Questions: Which brand(s) of dietary supplements should I purchase?
- For information about building a healthy dietary pattern
  - Dietary Guidelines for Americans

Glossary

absorption
: In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

Alzheimer's disease
: A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

blood vessel
: A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

cell
: The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

cognition
: The intellectual and mental ability to be aware, think, learn, imagine, remember, reason, have perceptions, and make judgments.

cognitive function
: Mental awareness and judgment.

control
: In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

cruciferous vegetable
: A type of vegetable including arugula, bok choy, broccoli, Brussels sprouts, cabbage, cauliflower, collards, kale, kohlrabi, mustard greens, radishes, rutabaga, turnips and watercress.

dairy food
: Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency
: An amount that is not enough; a shortage.

dementia
: Damaged brain function (thinking, learning, making decisions, remembering) that worsens over time. It disrupts activities of daily living such as bathing, dressing, and walking.

dietary fiber
: A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans
: Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

fortified
: When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gene
: The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

infant
: A child younger than 12 months old.

interaction
: A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

intravenous
: Into or within a vein, such as an intravenous injection.

liver
: A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

milligram
: mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral
: In nutrition, an inorganic substance found in the earth that is required to maintain health.

multivitamin/mineral dietary supplement
: MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nervous system
: The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

nutrient
: A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements
: ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

overweight
: An excess amount of body weight that includes muscle, bone, fat, and water. Overweight can be assessed by calculating the body mass index (BMI). (BMI is a number that estimates the amount of body fat on a person, based on weight and height. An adult with a BMI between 25 and 29.9 is considered overweight. Some people, such as bodybuilders or other athletes with a lot of muscle, can be overweight without having obesity. See: obesity.

pharmacist
: A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry
: Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription
: A written order from a health care provider for medicine, therapy, or tests.

prevent
: To stop from happening.

regulate
: To govern, make uniform, and bring under the control of a rule, principle, or legal system. In the United States, the FDA has the authority to regulate dietary supplements.

risk
: The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

supplement
: A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom
: A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

treat
: To care for a patient with a disease by using medicine, surgery, or other approaches.

upper limit
: UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture
: USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin
: A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

whole grain
: Unprocessed seeds of edible grasses, including brown rice, buckwheat, bulgur, millet, popcorn, oats, quinoa, whole-grain barley, whole rye, whole wheat, and wild rice. Grains that are ground, cracked, or flaked can be labeled whole grain if they have the same amount of bran, germ, and endosperm (the inner part of the seed kernel) as the intact grain. Whole grains are sources of iron, magnesium, selenium, B vitamins, and dietary fiber. Eating whole grains may help lower the risk of heart disease, obesity, and type 2 diabetes.

Updated: June 2, 2022